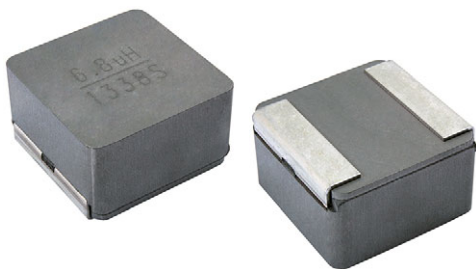




IHLP® Commercial Inductors, High Temperature (155 °C) Series



ADDITIONAL RESOURCES



FEATURES

- High temperature rating, up to 155 °C
- Shielded construction
- Excellent DC/DC energy storage up to 1 MHz to 2 MHz. Filter inductor applications up the SRF (see Standard Electrical Specifications table).
- Lowest DCR/μH, in this package size
- Handles high transient current spikes without saturation
- Ultra low buzz noise, due to composite construction
- IHLP design. PATENT(S): www.vishay.com/patents
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912



RoHS
COMPLIANT
HALOGEN
FREE
GREEN
(5-2008)

STANDARD ELECTRICAL SPECIFICATIONS

L ₀ INDUCTANCE ± 20 % AT 100 kHz, 0.25 V, 0 A (μH)	DCR TYP. 25 °C (mΩ)	DCR MAX. 25 °C (mΩ)	HEAT RATING CURRENT DC TYP. (A) ⁽¹⁾	SATURATION CURRENT DC TYP. (A) ⁽²⁾	SRF TYP. (MHz)
0.47	0.56	0.67	80.0	100.0	47.5
1.0	0.82	0.89	69.0	71.0	25.7
2.2	1.15	1.25	58.0	48.0	17.5
3.3	1.63	1.77	49.0	41.0	12.8
4.7	1.69	1.84	47.0	37.0	10.2
6.8	2.84	3.09	36.0	36.0	8.03
10	4.04	4.14	28.0	28.0	6.04
15	5.62	6.11	23.5	24.0	4.71
22	10.60	10.80	17.5	16.0	3.88
33	15.10	15.40	15.5	10.5	3.01
47	17.30	17.70	13.5	10.0	2.99
75	29.76	32.35	10.6	9.5	2.01
82	31.46	34.20	10.2	9.0	2.07
100	36.25	39.40	9.1	7.0	2.01

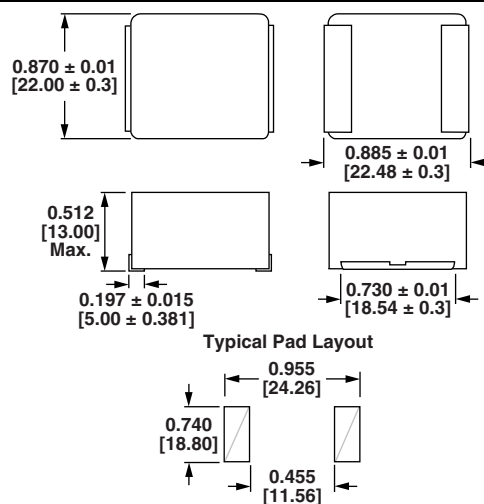
Notes

- All test data is referenced to 25 °C ambient
- Operating temperature range -55 °C to +155 °C
- The part temperature (ambient + temp. rise) should not exceed 155 °C under worst case operating conditions. Circuit design, component placement, PWB trace size and thickness, airflow and other cooling provisions all affect the part temperature. Part temperature should be verified in the end application
- Rated operating voltage (across inductor) = 75 V
- (1) DC current (A) that will cause an approximate ΔT of 40 °C
- (2) DC current (A) that will cause L₀ to drop approximately 20 %

APPLICATIONS

- Low profile, high current power supplies
- High current POL converters
- DC/DC converters in distributed power systems
- Servers
- Solar inverters
- Industrial lighting
- Industrial power supplies

DIMENSIONS in inches [millimeters]



DESCRIPTION

IHLP-8787MZ-51	100 μH	± 20 %	ER	e3
MODEL	INDUCTANCE VALUE	INDUCTANCE TOLERANCE	PACKAGE CODE	JEDEC® LEAD (Pb)-FREE STANDARD

GLOBAL PART NUMBER

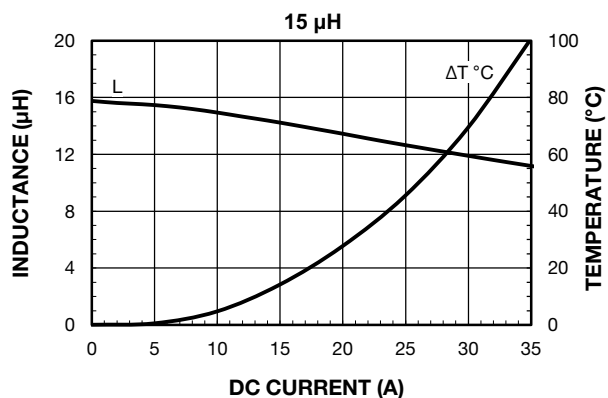
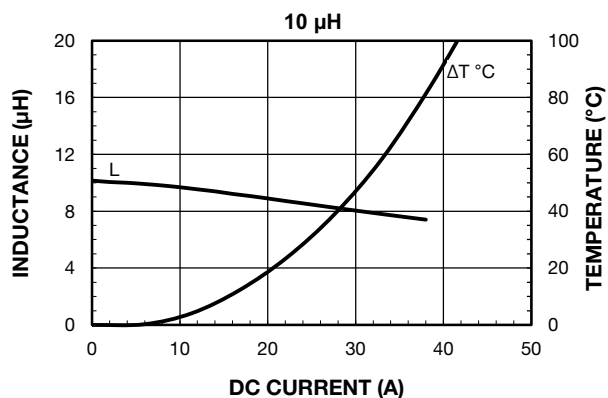
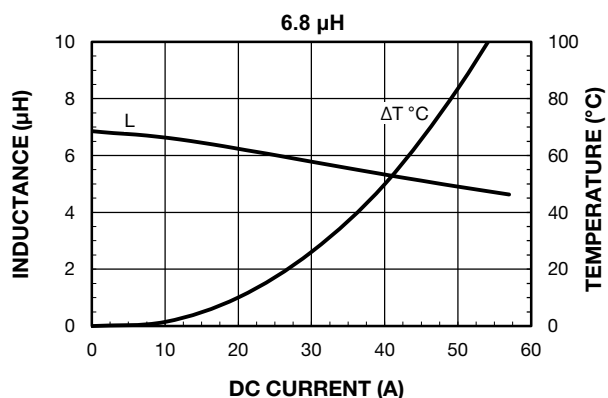
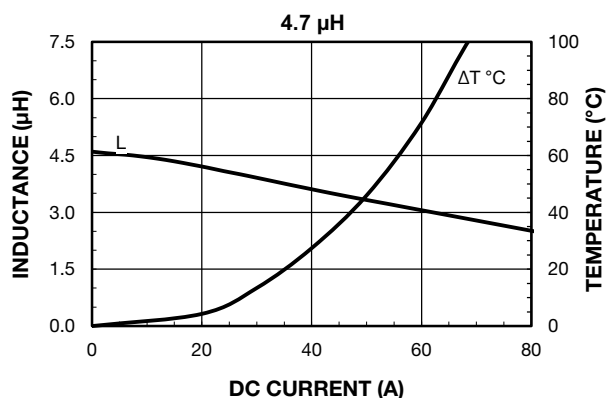
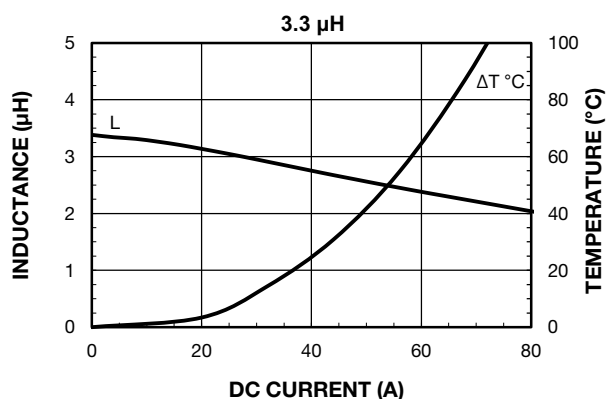
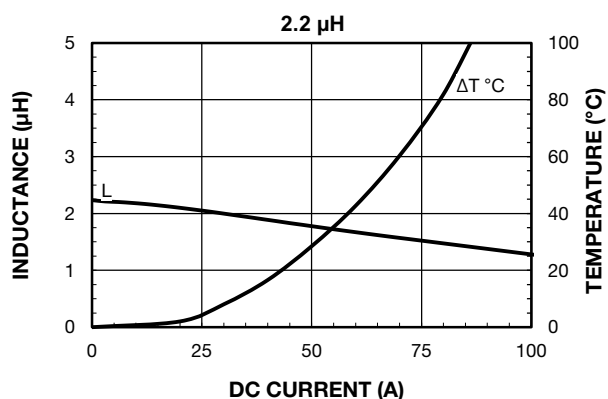
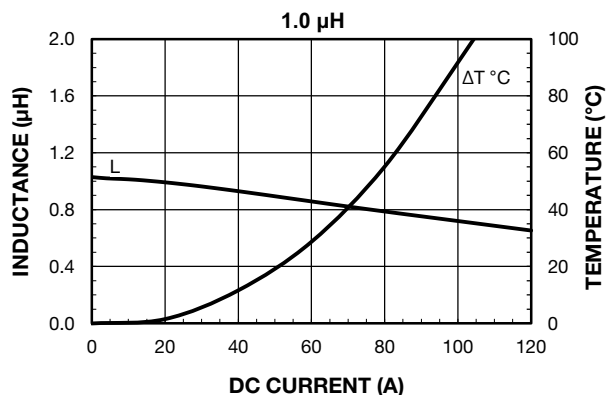
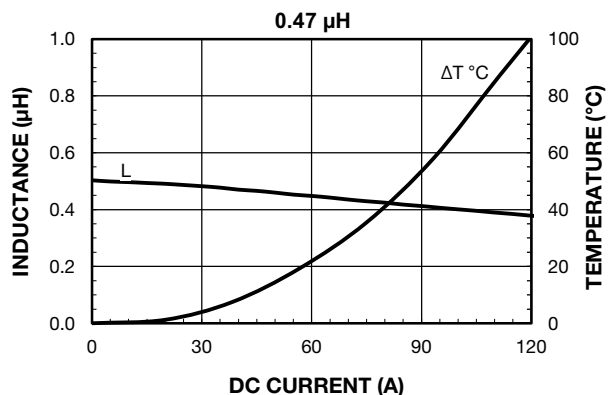
I	H	L	P	8	7	8	7	M	Z	E	R	1	0	1	M	5	1
PRODUCT FAMILY				SIZE						PACKAGE CODE		INDUCTANCE VALUE			TOL.	SERIES	

PATENT(S): www.vishay.com/patents

This Vishay product is protected by one or more United States and international patents.

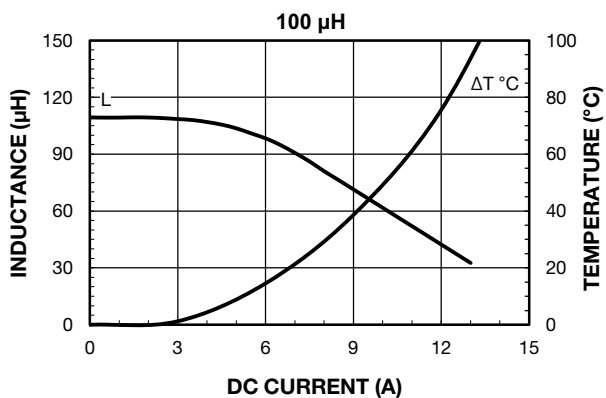
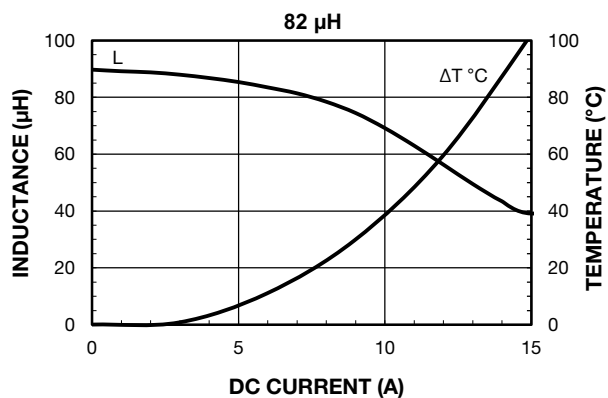
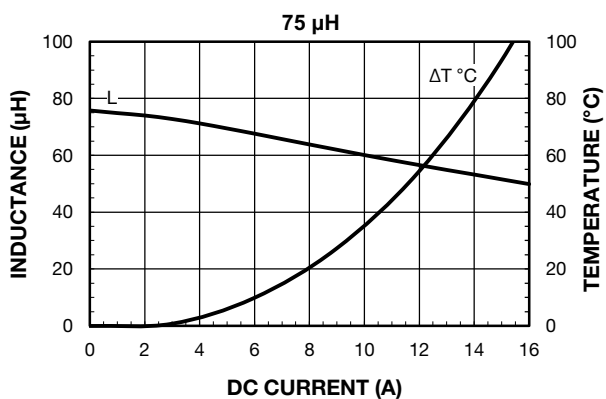
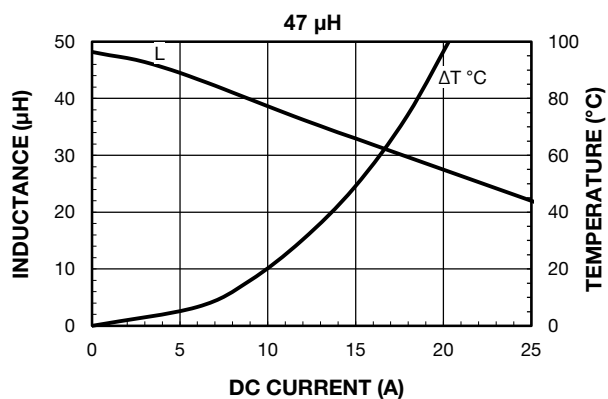
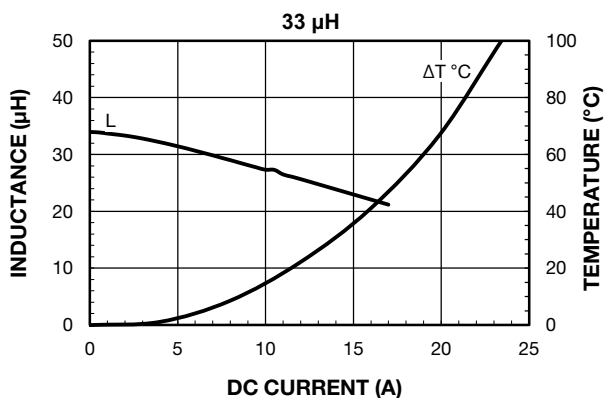
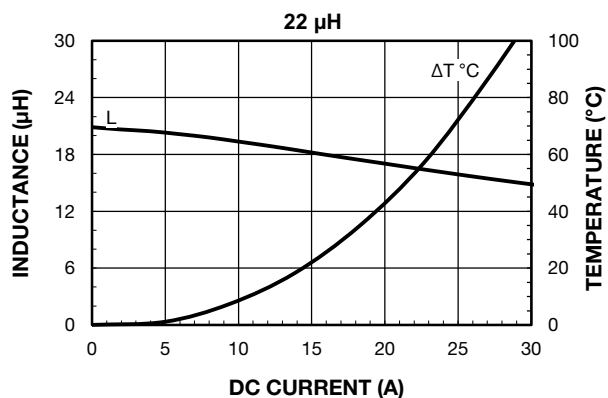


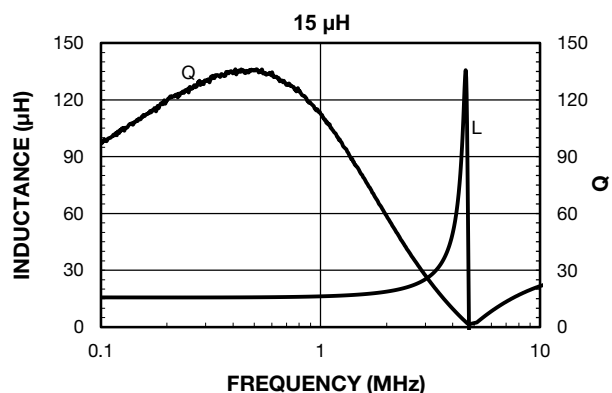
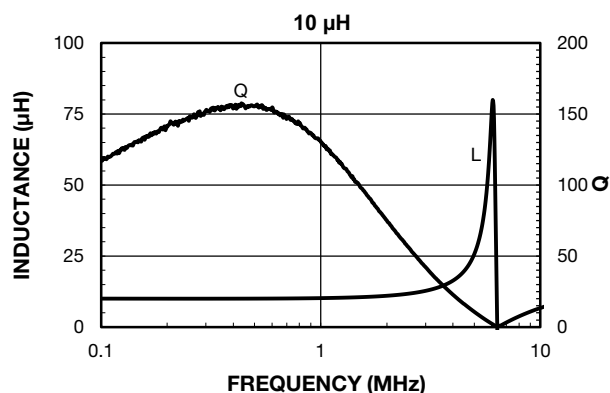
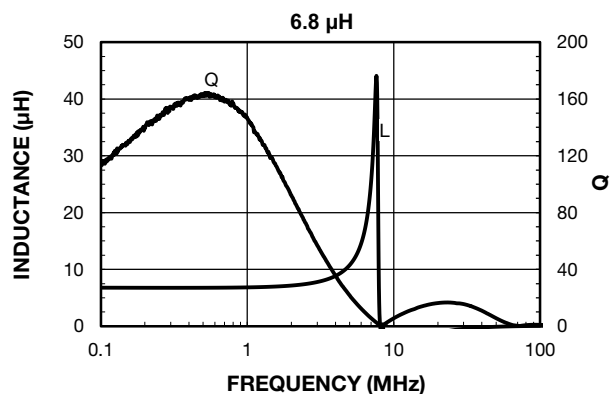
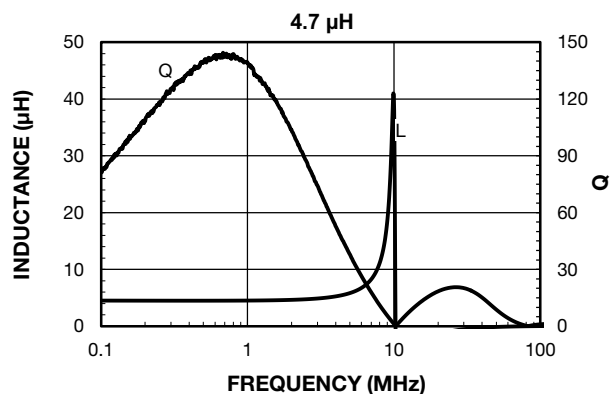
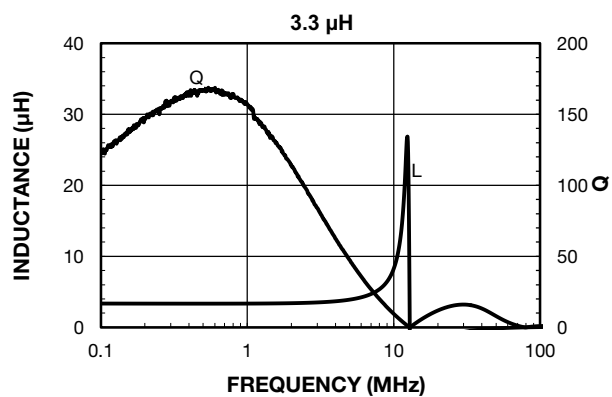
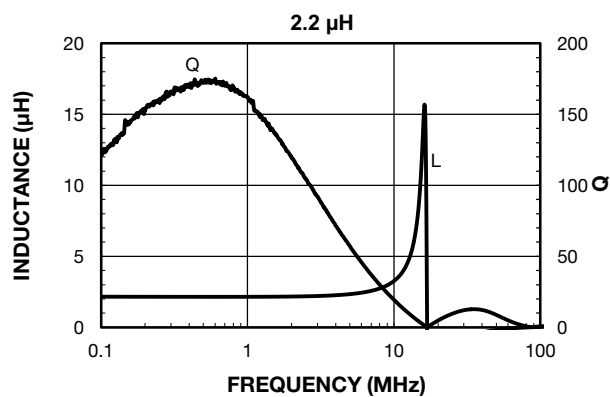
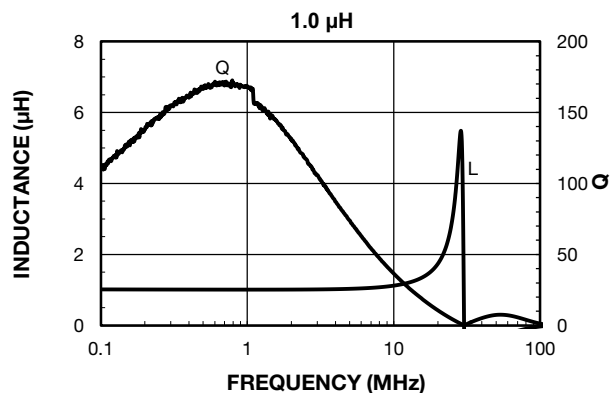
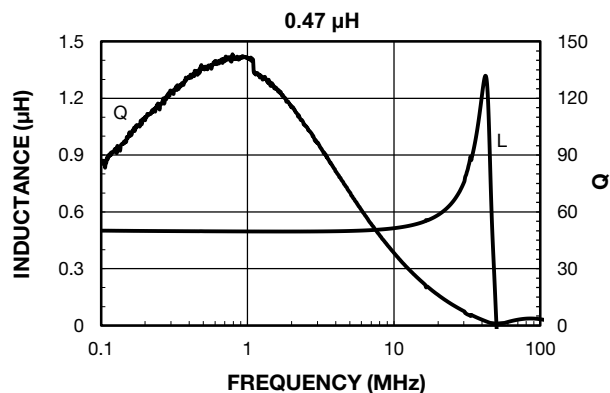
PERFORMANCE GRAPHS





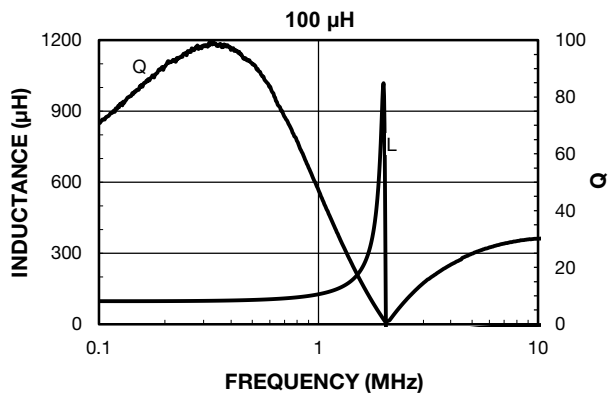
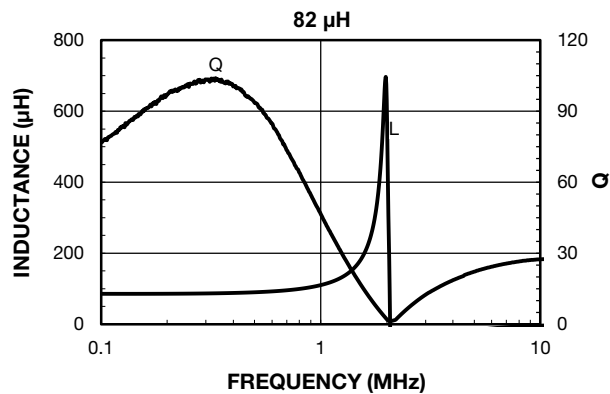
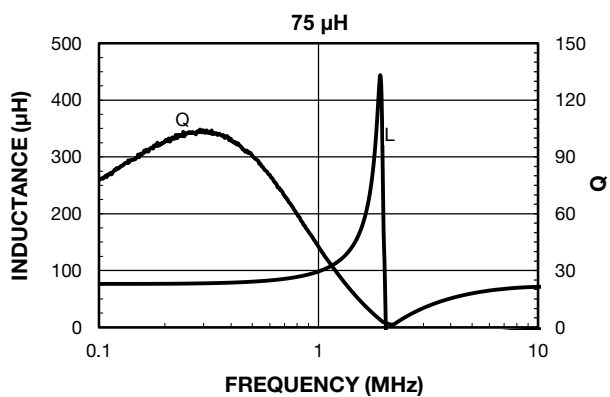
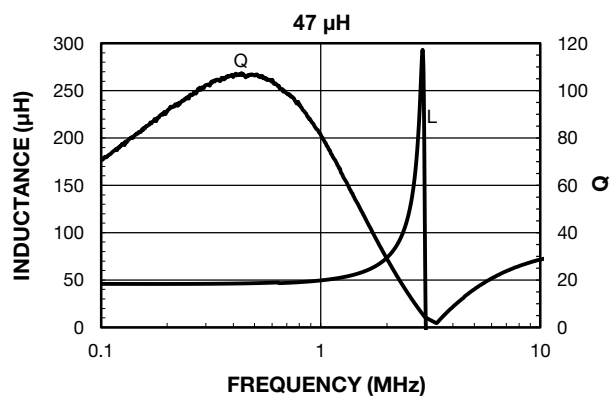
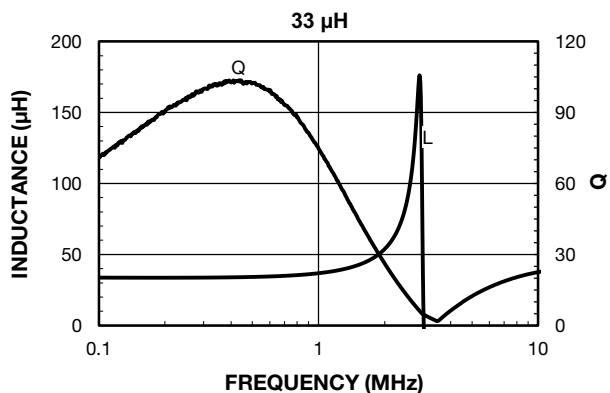
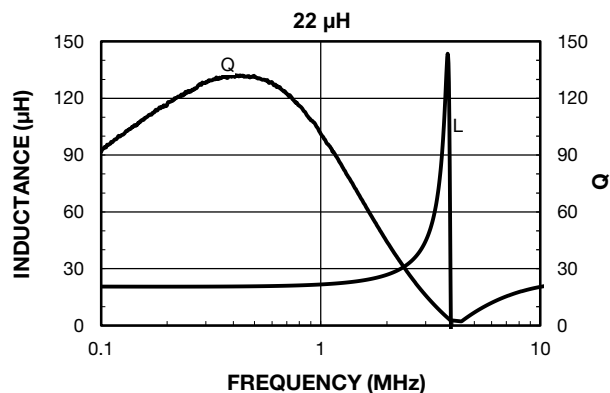
PERFORMANCE GRAPHS



PERFORMANCE GRAPHS: INDUCTANCE AND Q VS. FREQUENCY




PERFORMANCE GRAPHS: INDUCTANCE AND Q VS. FREQUENCY





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